Roots Of Polynomials

Chapter Overview

Solution 2

This chapter relates polynomial coefficients, such as a, b, c in $ax^2 + bx + c$, to its roots: values where the polynomial is zero.

1. Relationships for quadratics, cubics and quartics

Given $kx^2 + (k-3)x - 2 = 0$, find k when the roots sum to 4.

2. Expressions in the roots

Without finding the roots of $x^3 - 3x^2 - 3x + 1 = 0$, find the sum of their squares.

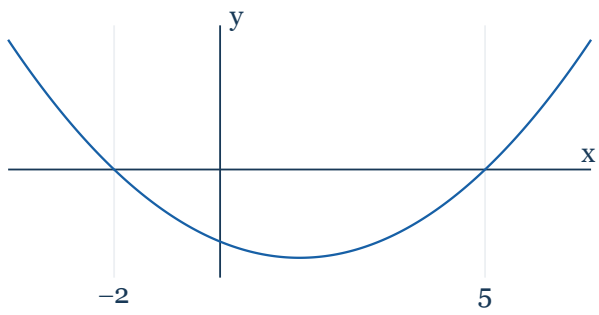
3. Linear transformations of roots

If $x^4 - 3x^3 + 15x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$, find an equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1, 2\delta + 1$.

Introduction

Solution 3

$p(x) = x^2 - 3x - 10$ has roots $-2, 5$.



We want to understand how roots relate to coefficients.

What do you notice about the sum $-2 + 5 = 3$?

It is the negative of the coefficient of x.

What do you notice about the product $(-2)(5) = -10$?

It equals the constant term.

At GCSE, factorising a monic quadratic used two numbers whose sum gives the middle coefficient and product the constant. The roots are the negatives of those factor constants.

Introduction

Solution 4

Let α, β be roots of $ax^2 + bx + c = 0$, $a \neq 0$.

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$
$$= a[x^2 - (\alpha + \beta)x + \alpha\beta]$$
$$= ax^2 - a(\alpha + \beta)x + a\alpha\beta.$$

Compare coefficients:

$b = -a(\alpha + \beta), \qquad c = a\alpha\beta.$

Quadratic roots

$\alpha + \beta = -b/a, \qquad \alpha\beta = c/a.$

Higher-degree Polynomials

Solution 5

There are similar relationships for higher degrees; always assume the leading coefficient is nonzero.

Quadratic: $ax^2 + bx + c$

Roots α, β.

$\alpha + \beta = -b/a, \quad \alpha\beta = c/a.$

Cubic: $ax^3 + bx^2 + cx + d$

$\alpha + \beta + \gamma = -b/a$

$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$

$\alpha\beta\gamma = -d/a$

Quartic: $ax^4 + bx^3 + cx^2 + dx + e$

$\alpha + \beta + \gamma + \delta = -b/a$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = c/a$

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d/a$

Spot the pattern. What is the product of all four roots?

$\alpha\beta\gamma\delta = e/a.$

For a quintic the product is $-f/a$; for a sextic it is g/a. Signs alternate. These are Vieta's formulae.

Back to roots of quadratics...

Solution 6

The roots of $2x^2 - 5x - 4 = 0$ are α, β. Without solving, find (a) their sum, (b) their product, (c) $1/\alpha + 1/\beta$, (d) $\alpha^2 + \beta^2$.

Use the sum and product to calculate these expressions without needing the individual roots. For the last two, rewrite in terms of $\alpha + \beta$ and $\alpha\beta$.

a $a = 2, b = -5, c = -4$; $\alpha + \beta = -b/a = 5/2.$

b $\alpha\beta = c/a = -2.$

c
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{5/2}{-2} = -5/4.$$

d
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (5/2)^2 + 4 = 41/4.$$

Further Example

Solution 7

The quadratic $ax^2 + bx + c = 0$ has roots $-3/2, 5/4$. Find integer a, b, c.

Initially choose $a = 1$; rescale to make all coefficients integers.

$\alpha + \beta = -3/2 + 5/4 = -1/4 = -b/a,$

$\alpha\beta = (-3/2)(5/4) = -15/8 = c/a.$

Taking $a = 1$ gives $b = 1/4, c = -15/8$:

$x^2 + x/4 - 15/8 = 0.$

Multiply by 8 without changing the roots:

$8x^2 + 2x - 15 = 0.$

$a = 8, b = 2, c = -15.$

Test Your Understanding

Solution 8

For $x^2 + 2x + 3 = 0$, find (a) the root sum, (b) the root product.

1a $a = 1, b = 2, c = 3$, so $\alpha + \beta = -2.$

1b

$\alpha\beta = 3.$

A quadratic has roots $2/3, 1/5$. Find integer coefficients a, b, c.

2

$\alpha + \beta = 13/15 = -b/a, \quad \alpha\beta = 2/15 = c/a.$

Choose $a = 1$: $x^2 - 13x/15 + 2/15 = 0.$

Multiply by 15:

$15x^2 - 13x + 2 = 0.$

$a = 15, b = -13, c = 2.$

Exercise 4A - Core Pure Book 1 Pages 56-57

Solution 9

8. $kx^2 + (k-3)x - 2 = 0$. Find k if its roots sum to 4.

8

$4 = -\dfrac{k-3}{k} = \dfrac{3-k}{k}.$

$3 - k = 4k \Rightarrow 3 = 5k \Rightarrow k = 3/5.$

9. $nx^2 - (16+n)x + 256 = 0$ has real roots $\alpha, -\alpha$. Find n.

9

The root sum is zero:

$0 = \dfrac{16+n}{n} \Rightarrow 16 + n = 0 \Rightarrow n = -16.$

The resulting roots are ± 4, so they are real.

Exercise 4A - Core Pure Book 1 Pages 56-57

Solution 10

10. The roots of $6x^2 + 36x + k = 0$ are reciprocals. Find k.

10

Their product is $\alpha(1/\alpha) = 1$.

$c/a = k/6 = 1 \Rightarrow k = 6.$

11. $mx^2 + 4x + 4m = 0$ has roots $k, 2k$. Find m, k.

11

$3k = -4/m, \qquad 2k^2 = 4m/m = 4.$

Thus $k = \pm\sqrt{2}$ and $m = -4/(3k)$.

Pairs:

$(k, m) = (\sqrt{2}, -2\sqrt{2}/3), \quad (-\sqrt{2}, 2\sqrt{2}/3).$

Exercise 4A - Core Pure Book 1 Pages 56-57

Solution 11

12. $ax^2 + 8x + c = 0$, with real a, c, has roots α, α^*.

a. Given $\operatorname{Re}\alpha = 2$, find a.

b. Given $\operatorname{Im}\alpha = 3$, find c.

a

Write $\alpha = p + qi$, so $\alpha + \alpha^* = 2p$ and $\alpha\alpha^* = p^2 + q^2$.

With $p = 2$: $4 = -8/a \Rightarrow a = -2$.

b

With $q = 3$: $13 = c/(-2) \Rightarrow c = -26$.

Exercise 4A - Core Pure Book 1 Pages 56-57

Solution 12

13. $4x^2 + px + q = 0$, with real p, q, has roots α, α^*.

a. Given $\operatorname{Re}\alpha = -3$, find p.

b. If $\operatorname{Im}\alpha \neq 0$, find the range of q.

a Write $\alpha = m + ni$. Then $2m = -p/4$. Since $m = -3$, $-6 = -p/4$, so $p = 24$.

b

$m^2 + n^2 = q/4$, so $q = 4(9 + n^2) = 36 + 4n^2$.

Since $n \neq 0$, $q > 36$.

Roots of Cubics

Solution 13

A cubic $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$, has three roots counting multiplicity. For real coefficients, either all roots are real or one is real and two are non-real conjugates.

$$\begin{aligned} & a(x-\alpha)(x-\beta)(x-\gamma) \\ &= ax^3 - a\alpha x^2 - a\beta x^2 - a\gamma x^2 \\ &\quad + a\alpha\beta x + a\beta\gamma x + a\gamma\alpha x - a\alpha\beta\gamma \\ &= ax^3 - a(\alpha+\beta+\gamma)x^2 \\ &\quad + a(\alpha\beta+\beta\gamma+\gamma\alpha)x - a\alpha\beta\gamma. \end{aligned}$$

Comparing coefficients:

$\alpha + \beta + \gamma = -b/a$

$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$

$\alpha\beta\gamma = -d/a.$

The sum rule matches the quadratic rule. For quadratics there is only one pair product. For cubics there is only one triple product, the product of all three roots.

Examples

Solution 14

The roots of $2x^3 + 3x^2 - 4x + 2 = 0$ are α, β, γ. Without solving, find (a) their sum, (b) their pair-product sum, (c) their product, (d) the sum of their reciprocals.

For the reciprocals, rewrite using the preceding symmetric expressions.

a

$a = 2, b = 3, c = -4, d = 2.$

$\alpha + \beta + \gamma = -3/2.$

b $\alpha\beta + \beta\gamma + \gamma\alpha = -4/2 = -2.$

c

$\alpha\beta\gamma = -2/2 = -1.$

d

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{-2}{-1} = 2.$$

Examples

A cubic has roots $1-2i, 1+2i, 2$. Find integer coefficients a, b, c, d.

As before, first choose a leading coefficient of 1.

Solution 15

$\alpha+\beta+\gamma = 4 = -b/a.$

$\alpha\beta+\beta\gamma+\gamma\alpha$

$= (1-2i)(1+2i) + 2(1+2i) + 2(1-2i)$

$= 5 + 4 = 9 = c/a.$

$\alpha\beta\gamma = (1-2i)(1+2i)2 = 10 = -d/a.$

Thus one choice is $a=1, b=-4, c=9, d=-10$, giving $x^3 - 4x^2 + 9x - 10 = 0$.

Exercise 4B - Core Pure Book 1 Pages 58-59

8. $2x^3 - kx^2 + 30x - 13 = 0$ has roots α, β, γ.

a. State the pair-product sum and product, and express k in terms of the roots.

b. Given $\alpha = 2-3i$, find k.

Solution 16

a

$\alpha\beta+\beta\gamma+\gamma\alpha = 15, \quad \alpha\beta\gamma = 13/2.$

The sum is $k/2$, hence $k = 2(\alpha+\beta+\gamma)$.

b

The conjugate root is $2+3i$. Their product is 13, so $13\gamma = 13/2 \Rightarrow \gamma = 1/2$.

$k = 2[(2-3i) + (2+3i) + 1/2] = 9.$

Exercise 4B Core Pure Book 1 Pages 58-59

Solution 17

9. $x^3 - mx + n = 0$ has roots $1, -4, \alpha$.

a. State, with a reason, whether α is real.

b. Find m, n, α.

a

It is real. A non-real root would require its distinct conjugate, in addition to the two given real roots, exceeding the degree.

b

The sum is zero: $1 - 4 + \alpha = 0 \Rightarrow \alpha = 3$.

Pair products:

$1(-4) + (-4)3 + 3(1) = -13 = -m$, so $m = 13$.

Product: $1(-4)3 = -12 = -n$, so $n = 12$.

Exercise 4B - Core Pure Book 1 Pages 58-59

Solution 18

10. $2x^3 - 10x^2 + 8x - k = 0$ has root $3 - i$.

a. Find the other roots.

b. Find k.

a

Another root is $3 + i$. The sum is $10/2 = 5$, so $6 + \gamma = 5 \Rightarrow \gamma = -1$.

The other roots are $3 + i, -1$.

b

$(3 - i)(3 + i)(-1) = 10(-1) = -10 = k/2$.

Hence $k = -20$.

Exercise 4B - Core Pure Book 1 Pages 58-59

Solution 19

11. $x^3 - 14x^2 + 56x - 64 = 0$ has roots $\alpha, k\alpha, k^2\alpha$, with real k. Find α, k.

11

From sum and pair products:

$\alpha(1 + k + k^2) = 14,$

$\alpha^2 k(1 + k + k^2) = 56.$

Divide the second by the first: $\alpha k = 4$. The product confirms $(\alpha k)^3 = 64$.

11

Substitute $\alpha = 4/k$ into the sum:

$\frac{4}{k}(1 + k + k^2) = 14 \Rightarrow 2k^2 - 5k + 2 = 0.$

$(2k - 1)(k - 2) = 0.$

Thus $(\alpha, k) = (2, 2)$ or $(8, 1/2)$. Both describe roots $2, 4, 8$.

Exercise 4B - Core Pure Book 1 Pages 58-59

Solution 20

12. $8x^3 + 12x^2 - cx + d = 0$ has roots $\alpha, \alpha/2, \alpha - 4$. Find α, c, d.

12

Root sum:

$\alpha + \alpha/2 + \alpha - 4 = -12/8 = -3/2 \Rightarrow \alpha = 1.$

Roots are $1, 1/2, -3$. Pair products:

$1/2 - 3/2 - 3 = -4 = -c/8 \Rightarrow c = 32.$

Product:

$-3/2 = -d/8 \Rightarrow d = 12.$

Challenge: Why the Expressions Are Real

Solution 21

α, β, γ are roots of a real-coefficient cubic. Considering all real/non-real possibilities, explain why (a) their sum, (b) their pair-product sum, (c) their product are real.

Start

If all roots are real, every expression is real.

Otherwise write the roots as $u+vi, u-vi, r$, with real u, v, r.

a $(u+vi)+(u-vi)+r = 2u+r \in \mathbb{R}.$

b

$(u+vi)(u-vi)+r(u-vi)+r(u+vi)$

$= u^2+v^2+2ur \in \mathbb{R}.$

c

$(u+vi)(u-vi)r = (u^2+v^2)r \in \mathbb{R}.$

Roots of Quartics

Solution 22

Quadratic: ax^2+bx+c

Roots α, β.

$\alpha+\beta = -b/a, \quad \alpha\beta = c/a.$

Cubic: ax^3+bx^2+cx+d

$\alpha+\beta+\gamma = -b/a$

$\alpha\beta+\beta\gamma+\gamma\alpha = c/a$

$\alpha\beta\gamma = -d/a$

Quartic: $ax^4+bx^3+cx^2+dx+e$

$\alpha+\beta+\gamma+\delta = -b/a$

$\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta = c/a$

$\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta = -d/a$

For the quartic, the product of all four roots is

$\alpha\beta\gamma\delta = e/a.$

Count each distinct pair and triple exactly once.

Example

Solution 23

$x^4 + 2x^3 + px^2 + qx - 60 = 0$, $p, q \in \mathbb{R}$, has roots $\alpha, \beta, \gamma, \delta$, with $\gamma = -2 + 4i$, $\delta = \gamma^*$.

a. Show $\alpha + \beta - 2 = 0$, $\alpha\beta + 3 = 0$.

b. Find all roots and p, q.

$\mathbb{C}$ denotes complex numbers. A notation such as $\Sigma\alpha\beta$ means the sum over all distinct pairs.

a

$\alpha + \beta + (-2 + 4i) + (-2 - 4i) = -2 \Rightarrow \alpha + \beta = 2.$

$\alpha\beta(-2 + 4i)(-2 - 4i) = -60$

$\Rightarrow 20\alpha\beta = -60 \Rightarrow \alpha\beta = -3.$

b

The remaining roots solve $t^2 - 2t - 3 = 0$, so are $3, -1$.

Pair products give

$p = \alpha\beta + (\alpha + \beta)(\gamma + \delta) + \gamma\delta$

$= -3 + 2(-4) + 20 = 9.$

Triple products give

$-q = \alpha\beta(\gamma + \delta) + (\alpha + \beta)\gamma\delta$

$= (-3)(-4) + 2(20) = 52 \Rightarrow q = -52.$

Expanded products:

$p = 3(-1) + 3(-2 + 4i) + 3(-2 - 4i)$

$+(-1)(-2 + 4i) + (-1)(-2 - 4i)$

$+(-2 + 4i)(-2 - 4i) = 9.$

$-q = 3(-1)(-2 + 4i) + 3(-1)(-2 - 4i)$

$+(-1)(-2 + 4i)(-2 - 4i)$

$+3(-2 + 4i)(-2 - 4i) = 52.$

Exercise 4C - Core Pure Book 1 Pages 60-61

Solution 24

8. $x^4 - 16x^3 + 86x^2 - 176x + 105 = 0$ has roots $\alpha, \alpha + k, \alpha + 2k, \alpha + 3k$, with real k. Solve the equation.

8

Root sum: $4\alpha + 6k = 16$, so $\alpha = 4 - 3k/2$.

The six distinct pair products are

$S_2 = \alpha(\alpha + k) + \alpha(\alpha + 2k) + \alpha(\alpha + 3k)$

$+(\alpha + k)(\alpha + 2k) + (\alpha + k)(\alpha + 3k)$

$+(\alpha + 2k)(\alpha + 3k)$

$= \alpha^2 + \alpha k + \alpha^2 + 2\alpha k + \alpha^2 + 3\alpha k$

$+\alpha^2 + 3\alpha k + 2k^2$

$+\alpha^2 + 4\alpha k + 3k^2 + \alpha^2 + 5\alpha k + 6k^2$

$= 6\alpha^2 + 18\alpha k + 11k^2.$

Hence

$6\alpha^2 + 18\alpha k + 11k^2 = 86.$

8

Substitute $\alpha = 4 - 3k/2$:

$6(4 - 3k/2)^2 + 18k(4 - 3k/2) + 11k^2 = 86$

$96 - 72k + 27k^2/2 + 72k - 27k^2 + 11k^2 = 86$

$-5k^2/2 = -10 \Rightarrow k = \pm 2.$

If $k = 2$, $\alpha = 1$, giving roots $1, 3, 5, 7$. If $k = -2$, $\alpha = 7$, giving the same roots in reverse.

Exercise 4C - Core Pure Book 1 Pages 60-61 — Solution 25

9. $3072x^4 - 2880x^3 + 840x^2 - 90x + 3 = 0$ has roots $\alpha, r\alpha, r^2\alpha, r^3\alpha$, with real r. Solve the equation.

1. Pair products

$S_2 = \alpha(r\alpha) + \alpha(r^2\alpha) + \alpha(r^3\alpha)$
$+(r\alpha)(r^2\alpha) + (r\alpha)(r^3\alpha)$
$+(r^2\alpha)(r^3\alpha)$
$= \alpha^2(r + r^2 + r^3 + r^3 + r^4 + r^5)$
$= \alpha^2 r(1 + r + 2r^2 + r^3 + r^4).$
$S_2 = 35/128.$

2. Triple products

$S_1 = \alpha(1 + r + r^2 + r^3) = 15/16.$
$S_3 = \alpha(r\alpha)(r^2\alpha) + \alpha(r\alpha)(r^3\alpha)$
$+\alpha(r^2\alpha)(r^3\alpha)$
$+(r\alpha)(r^2\alpha)(r^3\alpha)$
$= \alpha^3(r^3 + r^4 + r^5 + r^6)$
$= \alpha^3 r^3(1 + r + r^2 + r^3).$
$S_4 = \alpha(r\alpha)(r^2\alpha)(r^3\alpha)$
$= \alpha^4 r^6 = 1/1024.$
$S_3 = 15/512.$

3. Solve

$S_3/S_1 = \alpha^2 r^3 = 1/32$. Since $S_1 \neq 0$ and r is real, α is real; hence $r > 0$.

Put $t = r + 1/r \geq 2$. Then

$\frac{S_2}{\alpha^2 r^3}$
$= r^2 + r + 2 + r^{-1} + r^{-2}$
$= t^2 + t = 35/4.$
$(2t - 5)(2t + 7) = 0 \Rightarrow t = 5/2.$

Therefore $r = 2$ or $1/2$. The sum gives $\alpha = 1/16$ or $1/2$, respectively.

All roots are $1/16, 1/8, 1/4, 1/2$. Their sum and products verify the coefficients.

Sum check for $r = \alpha = 1/2$:

$\frac{1}{2}\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right)$
$= \frac{1}{2} \cdot \frac{15}{8} = \frac{15}{16}.$

Exercise 4C - Core Pure Book 1 Pages 60-61 — Solution 26

10. Three roots of $40x^4 + 90x^3 - 115x^2 + mx + n = 0$ are $1, -3, 1/2$.

a. Find the fourth root.

b. Find m, n.

a $1 - 3 + 1/2 + \delta = -90/40 = -9/4 \Rightarrow \delta = -3/4.$

b

Triple products:

$-3/2 + 9/4 - 3/8 + 9/8 = 3/2 = -m/40$

$\Rightarrow m = -60.$

Product:

$1(-3)(1/2)(-3/4) = 9/8 = n/40 \Rightarrow n = 45.$

Exercise 4C - Core Pure Book 1 Pages 60-61

Solution 27

11. $2x^4 - 34x^3 + 202x^2 + dx + e = 0$ has roots $\alpha, \alpha+1, 2\alpha+1, 3\alpha+1$.

a. Find α.

b. Find d, e.

a Sum: $7\alpha + 3 = 34/2 = 17$, hence $\alpha = 2$. The roots are $2, 3, 5, 7$.

b

Triple sum:

$2(3)(5) + 2(3)(7) + 2(5)(7) + 3(5)(7)$

$= 247 = -d/2.$

Thus $d = -494$. Product:

$2(3)(5)(7) = 210 = e/2 \Rightarrow e = 420.$

Exercise 4C - Core Pure Book 1 Pages 60-61

Solution 28

12. $4x^4 - 19x^3 + px^2 + qx + 10 = 0$, with real p, q, has roots $\alpha, \beta, 3+i, 3-i$.

a. Show $4\alpha + 4\beta + 5 = 0$, $4\alpha\beta - 1 = 0$.

b. Find all roots and p, q.

c. Plot the roots on an Argand diagram.

a

Sum: $\alpha + \beta + 6 = 19/4 \Rightarrow \alpha + \beta = -5/4$.

Product: $10\alpha\beta = 10/4 \Rightarrow \alpha\beta = 1/4$. These give the required equations.

b(ii)

Pair products:

$\frac{p}{4} = (-1)(-1/4) + (-1)(3+i) + (-1)(3-i)$

$+(-1/4)(3+i) + (-1/4)(3-i) + (3+i)(3-i)$

$= 1/4 - 6 - 3/2 + 10 = 11/4.$

Thus $p = 11$. Triple products:

$-\frac{q}{4} = (-1)(-1/4)(3+i) + (-1)(-1/4)(3-i)$

$+(-1)(3+i)(3-i) + (-1/4)(3+i)(3-i)$

$= 3/2 - 10 - 5/2 = -11.$

Thus $q = 44$.

b(i)

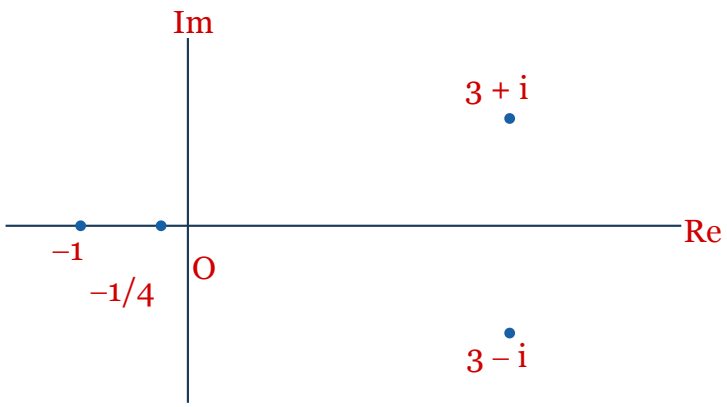
From $\alpha\beta = 1/4$, $\alpha = 1/(4\beta)$. Substitute into $4\alpha + 4\beta + 5 = 0$:

$1 + 4\beta^2 + 5\beta = 0$

$(4\beta + 1)(\beta + 1) = 0.$

The other roots are $-1, -1/4$. All four roots are $-1, -1/4, 3+i, 3-i$.

c



Expressions in the Roots

Solution 29

We can calculate symmetric expressions without finding individual roots. For a quadratic:

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta,$

$\dfrac{1}{\alpha}+\dfrac{1}{\beta}=\dfrac{\alpha+\beta}{\alpha\beta}\quad(\alpha\beta\neq 0).$

Sums of squares

For any number of roots, square the root sum and subtract twice the sum of distinct pair products.

Cubic:

$\alpha^2+\beta^2+\gamma^2=S_1^2-2S_2.$

Quartic:

$\alpha^2+\beta^2+\gamma^2+\delta^2=S_1^2-2S_2.$

Sums of cubes

Quadratic:

$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta).$

Cubic:

$\alpha^3+\beta^3+\gamma^3=S_1^3-3S_1S_2+3S_3.$

Here S_j sums distinct products of j roots. These results may be used without proof.

The same symmetric-sum pattern extends to higher degrees. A quartic sum-of-cubes formula is not required in this chapter.

Example

Solution 30

Three roots satisfy $\alpha\beta\gamma=4$, $\alpha\beta+\beta\gamma+\gamma\alpha=-5$, $\alpha+\beta+\gamma=3$. Find $(\alpha+3)(\beta+3)(\gamma+3)$.

Expand and group into the known symmetric expressions.

$(\alpha+3)(\beta+3)(\gamma+3)$

$=\alpha\beta\gamma+3\alpha\beta+3\beta\gamma+3\gamma\alpha+9\alpha+9\beta+9\gamma+27$

$=\alpha\beta\gamma+3(\alpha\beta+\beta\gamma+\gamma\alpha)+9(\alpha+\beta+\gamma)+27$

$=4+3(-5)+9(3)+27=43.$

Exercise 4D - Core Pure Book 1 Pages 63-64 Solution 31

11. The roots of $2x^3 + 4x^2 + 7 = 0$ are α, β, γ.

a. State their sum, pair-product sum and product.

b. Find (i) $\alpha^2 + \beta^2 + \gamma^2$, (ii) $\alpha^3\beta^3\gamma^3$, (iii) $(\alpha + 2)(\beta + 2)(\gamma + 2)$.

a

$S_1 = -2, S_2 = 0, S_3 = -7/2.$

b(i) $S_1^2 - 2S_2 = (-2)^2 = 4.$

b(ii)

$(\alpha\beta\gamma)^3 = (-7/2)^3 = -343/8.$

b(iii)

$S_3 + 2S_2 + 4S_1 + 8 = -7/2 + 0 - 8 + 8 = -7/2.$

Exercise 4D - Core Pure Book 1 Pages 63-64 Solution 32

12. Prove

$$\alpha^3 + \beta^3 + \gamma^3 = S_1^3 - 3S_1S_2 + 3S_3,$$

where $S_1 = \alpha + \beta + \gamma$, $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha$, $S_3 = \alpha\beta\gamma$.

12

Let $T = \alpha(\beta^2 + \gamma^2) + \beta(\alpha^2 + \gamma^2) + \gamma(\alpha^2 + \beta^2)$.

Expanding $S_1(\alpha^2 + \beta^2 + \gamma^2 + 2S_2)$ gives

$S_1^3 = \alpha^3 + \beta^3 + \gamma^3 + T + 2S_1S_2.$

12

Expand S_1S_2:

$\alpha^2\beta + \alpha\beta^2 + \alpha^2\gamma + \alpha\gamma^2 + \beta^2\gamma + \beta\gamma^2 + 3\alpha\beta\gamma$

$= T + 3S_3.$

Thus $T = S_1S_2 - 3S_3$. Substitution gives

$S_1^3 = \alpha^3 + \beta^3 + \gamma^3 + 3S_1S_2 - 3S_3.$

Rearrange to obtain the required formula.

Exercise 4D - Core Pure Book 1 Pages 63-64

13. $3x^3 - px + 11 = 0$ has roots α, β, γ.

a. If the pair-product sum is 4, find p.

b. State the root sum and product.

c. Find $(3-\alpha)(3-\beta)(3-\gamma)$.

Solution 33

a $-p/3 = 4 \Rightarrow p = -12.$

b
$S_1 = 0, S_3 = -11/3.$

c
$(3-\alpha)(3-\beta)(3-\gamma) = 27 - 9S_1 + 3S_2 - S_3$
$= 27 - 0 + 12 + 11/3 = 128/3.$

Exercise 4D - Core Pure Book 1 Pages 63-64

14. $x^4 + 2x^2 - x + 3 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. State S_1, S_2, S_3, S_4.

b. Find (i) the sum of reciprocals, (ii) the sum of squares, (iii) $(\alpha+1)(\beta+1)(\gamma+1)(\delta+1)$.

Solution 34

a
$S_1 = 0, S_2 = 2, S_3 = 1, S_4 = 3.$

b(i) $1/\alpha + 1/\beta + 1/\gamma + 1/\delta = S_3/S_4 = 1/3.$

b(ii)
$S_1^2 - 2S_2 = 0 - 4 = -4.$

b(iii)
Expanding by numbers of chosen roots:
$S_4 + S_3 + S_2 + S_1 + 1 = 3 + 1 + 2 + 0 + 1 = 7.$

Exercise 4D - Core Pure Book 1 Pages 63-64 — Solution 35

15. $ax^4 + 3x^3 + 2x^2 + x - 6 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. If their product is -3, find a.

b. State S_1, S_2, S_3.

c. Find the sum of reciprocals.

a $-6/a = -3 \Rightarrow a = 2.$

b
$S_1 = -3/2, S_2 = 2/2 = 1, S_3 = -1/2.$

c
$S_3/S_4 = (-1/2)/(-3) = 1/6.$

Exercise 4D - Core Pure Book 1 Pages 63-64 — Solution 36

16. Prove that for four roots,
$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = S_1^2 - 2S_2.$

16
Expand the square:

$(\alpha + \beta + \gamma + \delta)^2 = \alpha^2 + \beta^2 + \gamma^2 + \delta^2$
$+2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta).$

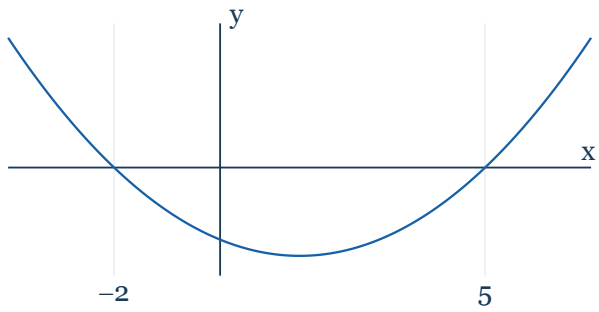
The final bracket is S_2, so rearrangement proves the result.

Linear Transformations of Roots

Solution 37

Original polynomial:

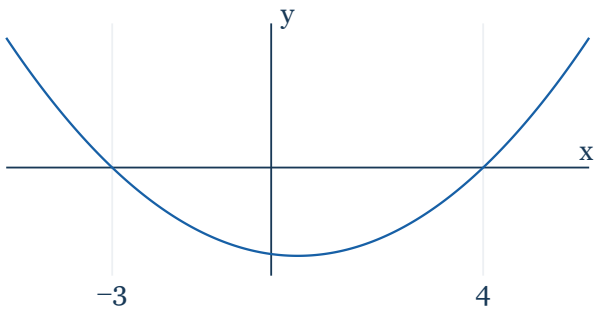
$p(x) = (x+2)(x-5) = x^2 - 3x - 10.$



How do the coefficients change when each root decreases by 1?

Transform each root by $\alpha \mapsto \alpha - 1$:

$(x+3)(x-4) = x^2 - x - 12.$



In factorised form, replace each root by one less. The effect on expanded coefficients is less immediate.

Continuing this example

Solution 38

Without finding the roots α, β of $x^2 - 3x - 10 = 0$, find an equation with roots $\alpha - 1, \beta - 1$.

Method 1: coefficients

Method 1

Original sum 3, product -10.

New sum: $(\alpha - 1) + (\beta - 1) = 3 - 2 = 1.$

New product:

$(\alpha - 1)(\beta - 1) = \alpha\beta - \alpha - \beta + 1$

$= -10 - 3 + 1 = -12.$

Hence $w^2 - w - 12 = 0.$

Method 2: substitution

Use a new variable for the transformed root.

Method 2

Let $w = x - 1$, so $x = w + 1$.

$(w+1)^2 - 3(w+1) - 10 = 0$

$w^2 + 2w + 1 - 3w - 3 - 10 = 0$

$w^2 - w - 12 = 0.$

This is the familiar function transformation: translating roots left by 1 means using $f(w+1)$. Substitution is generally the simpler method.

Quartic Example

Solution 39

The roots of $x^4 - 3x^3 + 15x + 1 = 0$ are $\alpha, \beta, \gamma, \delta$. Find an equation whose roots are twice each original root plus 1.

Method 1: symmetric sums

Method 1

Original $(S_1, S_2, S_3, S_4) = (3, 0, -15, 1)$.

New sums:

$T_1 = 2S_1 + 4 = 10,$

$T_2 = 4S_2 + 6S_1 + 6 = 24,$

$T_3 = 8S_3 + 8S_2 + 6S_1 + 4 = -98,$

$T_4 = 16S_4 + 8S_3 + 4S_2 + 2S_1 + 1 = -97.$

Hence $w^4 - 10w^3 + 24w^2 + 98w - 97 = 0.$

Method 2: substitution (recommended)

Method 2

Let $w = 2x + 1$, so $x = (w - 1)/2$.

$$\left(\frac{w-1}{2}\right)^4 - 3\left(\frac{w-1}{2}\right)^3 + 15\frac{w-1}{2} + 1 = 0.$$

Multiply by 16:

$$(w-1)^4 - 6(w-1)^3 + 120(w-1) + 16 = 0.$$

Expand and collect:

$$w^4 - 10w^3 + 24w^2 + 98w - 97 = 0.$$

Test Your Understanding

Solution 40

$x^3 - 2x^2 + 4 = 0$ has roots α, β, γ. Find an equation with roots $3\alpha - 1, 3\beta - 1, 3\gamma - 1$.

Method 1: symmetric sums

Method 1

Original $(S_1, S_2, S_3) = (2, 0, -4)$.

New sum $T_1 = 3S_1 - 3 = 3.$

New pair sum $T_2 = 9S_2 - 6S_1 + 3 = -9.$

New product $T_3 = 27S_3 - 9S_2 + 3S_1 - 1 = -103.$

Hence $w^3 - 3w^2 - 9w + 103 = 0.$

Method 2: substitution

Method 2

Let $w = 3x - 1$, so $x = (w + 1)/3$.

$$\left(\frac{w+1}{3}\right)^3 - 2\left(\frac{w+1}{3}\right)^2 + 4 = 0.$$

Multiply by 27:

$$(w+1)^3 - 6(w+1)^2 + 108 = 0$$

$$w^3 + 3w^2 + 3w + 1 - 6w^2 - 12w - 6 + 108 = 0$$

$$w^3 - 3w^2 - 9w + 103 = 0.$$

Exercise 4E - Core Pure Book 1 Pages 66-67 Solution 41

3. Without solving $x^3 - 3x^2 + 4x - 7 = 0$, find an equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$, in the form $aw^3 + bw^2 + cw + d = 0$ with integer coefficients.

3

Let $w = 2x + 1$, so $x = (w - 1)/2$.

$$\left(\frac{w-1}{2}\right)^3 - 3\left(\frac{w-1}{2}\right)^2 + 4\frac{w-1}{2} - 7 = 0.$$

Multiply by 8:

$$(w-1)^3 - 6(w-1)^2 + 16(w-1) - 56 = 0$$

$$w^3 - 3w^2 + 3w - 1 - 6w^2 + 12w - 6 + 16w - 16 - 56 = 0$$

$$w^3 - 9w^2 + 31w - 79 = 0.$$

Exercise 4E - Core Pure Book 1 Pages 66-67 Solution 42

4. Without solving $x^3 + 4x^2 - 4x + 2 = 0$, find an equation with roots $2\alpha - 1, 2\beta - 1, 2\gamma - 1$, in the form $w^3 + pw^2 + qw + r = 0$ with integers p, q, r.

4

Let $w = 2x - 1$, so $x = (w + 1)/2$. Substitute:

$$\left(\frac{w+1}{2}\right)^3 + 4\left(\frac{w+1}{2}\right)^2 - 4\left(\frac{w+1}{2}\right) + 2 = 0.$$

Multiply by 8:

$$(w+1)^3 + 8(w+1)^2 - 16(w+1) + 16 = 0$$

$$w^3 + 3w^2 + 3w + 1 + 8(w^2 + 2w + 1) - 16w - 16 + 16 = 0.$$

$$w^3 + 3w^2 + 3w + 1 + 8w^2 + 16w + 8 - 16w - 16 + 16 = 0$$

$$w^3 + 11w^2 + 3w + 9 = 0.$$

Exercise 4E - Core Pure Book 1 Pages 66-67 Solution 43

5. Without solving $3x^3 - x^2 + 2x - 5 = 0$, find an equation with roots $3\alpha + 1, 3\beta + 1, 3\gamma + 1$, with integer coefficients.

5

Let $w = 3x + 1$, so $x = (w - 1)/3$. Substitute:

$$3\left(\frac{w-1}{3}\right)^3 - \left(\frac{w-1}{3}\right)^2 + 2\left(\frac{w-1}{3}\right) - 5 = 0.$$

Multiply by 27:

$$3(w-1)^3 - 3(w-1)^2 + 18(w-1) - 135 = 0$$

$$3(w^3 - 3w^2 + 3w - 1) - 3(w^2 - 2w + 1) + 18w - 18 - 135 = 0.$$

$$3w^3 - 9w^2 + 9w - 3 - 3w^2 + 6w - 3 + 18w - 18 - 135 = 0$$

$$3w^3 - 12w^2 + 33w - 159 = 0.$$

Divide by 3:

$$w^3 - 4w^2 + 11w - 53 = 0.$$

Exercise 4E - Core Pure Book 1 Pages 66-67 Solution 44

6. $2x^4 + 4x^3 - 5x^2 + 2x - 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find integer-coefficient equations whose roots are (a) three times the original roots, (b) each original root minus 1.

a

Let $w = 3x$, so $x = w/3$.

$$2(w/3)^4 + 4(w/3)^3 - 5(w/3)^2 + 2w/3 - 1 = 0.$$

$$\frac{2w^4}{81} + \frac{4w^3}{27} - \frac{5w^2}{9} + \frac{2w}{3} - 1 = 0.$$

Multiply by 81:

$$2w^4 + 12w^3 - 45w^2 + 54w - 81 = 0.$$

b

Let $w = x - 1$, so $x = w + 1$.

$$2(w+1)^4 + 4(w+1)^3$$

$$-5(w+1)^2 + 2(w+1) - 1 = 0.$$

Expand:

$$2(w^4 + 4w^3 + 6w^2 + 4w + 1)$$

$$+4(w^3 + 3w^2 + 3w + 1)$$

$$-5(w^2 + 2w + 1) + 2w + 1 = 0.$$

Collect terms:

$$2w^4 + 8w^3 + 12w^2 + 8w + 2$$

$$+4w^3 + 12w^2 + 12w + 4$$

$$-5w^2 - 10w - 5 + 2w + 2 - 1 = 0.$$

$$2w^4 + 12w^3 + 19w^2 + 12w + 2 = 0.$$

7. Without solving $x^4 + 2x^3 - 3x^2 + 4x + 5 = 0$, find integer-coefficient equations with roots (a) twice each original root, (b) each original root minus 2.

a

Let $w = 2x$, so $x = w/2$.

$(w/2)^4 + 2(w/2)^3 - 3(w/2)^2 + 4w/2 + 5 = 0.$

$\frac{w^4}{16} + \frac{2w^3}{8} - \frac{3w^2}{4} + 2w + 5 = 0.$

Multiply by 16:

$w^4 + 4w^3 - 12w^2 + 32w + 80 = 0.$

b

Let $w = x - 2$, so $x = w + 2$.

$(w+2)^4 + 2(w+2)^3$

$-3(w+2)^2 + 4(w+2) + 5 = 0.$

$w^4 + 8w^3 + 24w^2 + 32w + 16$

$+2w^3 + 12w^2 + 24w + 16$

$-3w^2 - 12w - 12 + 4w + 8 + 5 = 0.$

$w^4 + 8w^3 + 24w^2 + 32w + 16$

$\quad +2(w^3 + 6w^2 + 12w + 8)$

$-3(w^2 + 4w + 4) + 4w + 8 + 5 = 0.$

$w^4 + 10w^3 + 33w^2 + 48w + 33 = 0.$

8. Without solving $3x^4 + 5x^3 - 4x^2 - 3x + 1 = 0$, find integer-coefficient equations with roots (a) three times each original root, (b) each original root plus 1.

a

Let $w = 3x$, so $x = w/3$.

$3(w/3)^4 + 5(w/3)^3 - 4(w/3)^2 - w + 1 = 0.$

$\frac{3w^4}{81} + \frac{5w^3}{27} - \frac{4w^2}{9} - w + 1 = 0.$

Multiply by 81:

$3w^4 + 15w^3 - 36w^2 - 81w + 81 = 0.$

Divide by 3:

$w^4 + 5w^3 - 12w^2 - 27w + 27 = 0.$

b

Let $w = x + 1$, so $x = w - 1$.

$3(w-1)^4 + 5(w-1)^3$

$-4(w-1)^2 - 3(w-1) + 1 = 0.$

$3w^4 - 12w^3 + 18w^2 - 12w + 3$

$+5w^3 - 15w^2 + 15w - 5$

$-4w^2 + 8w - 4 - 3w + 3 + 1 = 0.$

$3(w^4 - 4w^3 + 6w^2 - 4w + 1)$

$\quad +5(w^3 - 3w^2 + 3w - 1)$

$-4(w^2 - 2w + 1) - 3w + 3 + 1 = 0.$

$3w^4 - 7w^3 - w^2 + 8w - 2 = 0.$

Extension

Solution 47

The quartic $2x^4 - 3x^3 + x^2 - 2x - 6 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. Find an integer-coefficient equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1, 2\delta + 1$.

a

Let $w = 2x + 1$, so $x = (w - 1)/2$. Substitute:

$$2\left(\frac{w-1}{2}\right)^4 - 3\left(\frac{w-1}{2}\right)^3 + \left(\frac{w-1}{2}\right)^2 - 2\left(\frac{w-1}{2}\right) - 6 = 0.$$

Multiply by 8:

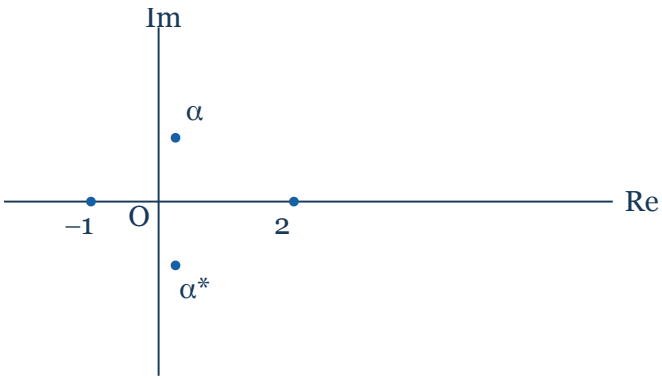
$$(w-1)^4 - 3(w-1)^3 + 2(w-1)^2 - 8(w-1) - 48 = 0$$

$$w^4 - 4w^3 + 6w^2 - 4w + 1 - 3w^3 + 9w^2 - 9w + 3 + 2w^2 - 4w + 2 - 8w + 8 - 48 = 0$$

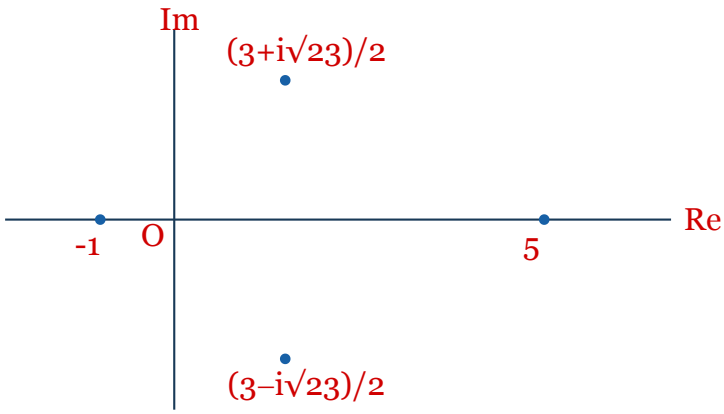
$$w^4 - 4w^3 + 6w^2 - 4w + 1 - 3(w^3 - 3w^2 + 3w - 1) + 2(w^2 - 2w + 1) - 8w + 8 - 48 = 0.$$

$$w^4 - 7w^3 + 17w^2 - 25w - 34 = 0.$$

b. The diagram shows the original roots. Sketch the roots after the transformation.



b



Why These Relationships Are Useful

Solution 48

What do sums and products tell us?

They connect the geometry of roots with polynomial coefficients. The mean of n roots is S_1/n; on an Argand diagram this is the centroid of their points, counting multiplicity.

For $z^k = a + ib$, integer $k \geq 2$, the sum of the roots is zero because the coefficient of z^{k-1} is zero.

Why is this useful?

If one root is missing, subtract the known roots from $-b/a$. If a coefficient is unknown, use information about the roots to determine it. These relationships strengthen our ability to reason about polynomials.

Why sums of squares or reciprocals?

They let us evaluate expressions without solving for individual roots, and form equations for transformed roots. This algebraic structure remains useful even when no immediate physical application is given.